

MANAGEMENT OF BURNS

Introduction

Burn is one of the most severe injuries a person can suffer. In India burns are more common in poor people due to over crowding, open stove cooking, floor level cooking, and poor house planning. In fact burn injury can be classified as a chronic disease as it requires long term management, supervised rehabilitation and psychosocial support.

Definition:

Burn is a tissue injury that results from thermal application (hot / cold) and from application of physical or chemical energy. Burns are caused by dry heat, moist heat, cold injury, chemical burns, electrical burns, ionizing radiation, and friction.

INCIDENCE OF BURNS IN OUR COUNTRY

India has a high incidence of burn injuries but the exact figures of incidence in India are not available. Largest group of patients suffering from burns are children and young women.

DIAGNOSTIC CRITERIA

Diagnosis of burns is straight forward and history gives clue to the causative agent.

A. History: This should aim at evaluating:

- Time of burn
- Possibility of being a medicolegal case.
- AMPLE history
 - Allergies
 - Medications
 - Past medical history
 - Last meal
 - Events and Environment of burn injury.

B. Physical examination and evaluation of burn wound.

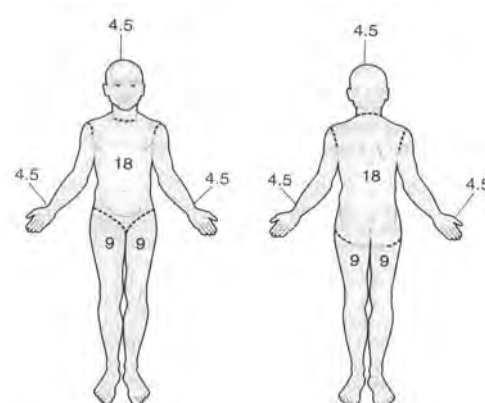
(i) Depth of burn

- I Degree burns : Minor epithelial damage, present as redness, clinically insignificant.
- II Degree burns : Partial thickness burns & scalds. These burns are painful
- III Degree burns : Full thickness burns, usually painless

(ii) Calculation of Surface area of burn.

Wallace's Rule of Nine is useful in adults. In children, Lund & Brewder's Chart can be used.

A rough guide is - surface area of palm is approximately equal to 1% of Total Body Surface Area (TBSA).



ADMISSION CRITERIA & CRITERIA FOR REFERRAL

1. 100 or more bedded hospital or a burn centre

All major burns like:

- **Partial** thickness burns of **>25%** of TBSA in **adults**
- **>20%** burns in **children below 10 yrs** and **adults >50 yrs.**
- **Full** thickness burns **>10% TBSA**
- Burns of face, eyes, ears, and perineum
- Chemical burns, electrical burns, inhalational injury, and underlying debilitating illness

2. 30-100 bedded community health centre hospital

Moderate burns like burns >15% TBSA (excluding burns described above).

MANAGEMENT

Thorough examination of burn patient for the presence of other injuries is essential as per the ATLS Primary survey protocol. If any life threatening condition found, it may be attended to simultaneously. The management specific to major burns is:

Airway

Airway burns, smoke inhalation syndrome, severe burns of face & neck, laryngeal edema may need intubation or tracheostomy.

O₂ inhalations:

Started at 8-12 lit/min in patients having dyspnea, major burns, airway burns, and smoke inhalation syndrome.

Fluid resuscitation

Insertion of large bore cannulae / cut downs are necessary. Parkland formula is to be used for calculation of fluid requirement in first 24 hrs i.e. 4ml/kg/% of burn. The fluid of choice is Ringers lactate. 50% of the fluid is to be transfused **in first 8 hrs from the time of burns** and the remaining 50% in next 16 hrs. Adequate resuscitation is monitored by vital parameters and urine output of 0.5 - 1ml /kg/hr. If the urine output falls below 0.5ml/kg/hr a bolus of 10ml/kg body weight can be given. If the urine output is in excess of 2ml/kg /hr the rate of infusion should be reduced.

Next 24 hrs:

Total volume = ½ of first day.

Colloids (0.5 ml / kg / %) and 5% glucose or isotonic glucose saline to make up the rest.

Catheterization

Patients with burns above 20% are to be catheterized.

Pain relief

- Morphine is the drug of choice except in respiratory burns.
 - Dose in adults is 10mg IV in dilute form given slowly.
 - In children, dose is 1mg per year of age (0.1-0.3mg/Kg).
- Alternate drugs for analgesia are Pethidine or Pentazocine in appropriate dosage.

Nasogastric tube insertion

In patients with more than 20% burns placing nasogastric tube and evacuation of fluid from stomach prevents vomiting and aspiration. It also helps in starting early feeds.

Antibiotics

- Antibiotics are not generally indicated in early post burn period.

- Use of broad spectrum antibiotics lead to colonization of wound by resistant organisms.
- However Ceftazidime + Amikacin + Metronidazole can be used after 48hrs pending sensitivity of organism isolated from the wound.

H2-receptor antagonists

- Acute upper GI erosions and ulcers (Curling ulcer) may occur in patients with severe burn injuries. The common clinical finding in such patients is painless GI hemorrhage. Treatment of acute stress ulceration is principally preventive.
- H2-receptor antagonists inhibit gastric acid secretion; this prophylaxis against acute stress ulceration is initiated immediately after admission. Commonly used drug is Inj. Ranitidine 50 mg X 8th hourly.

Tetanus prophylaxis

Inj. TT 0.5cc IM stat.

Isolation of the patient

As far as possible isolate burn patient from other patients, if possible in a separate ward or cubicle.

Care of burn wound

All sterile precautions need to be taken while dressing a burn wound.

- Clean wound with dilute Savlon /normal saline /Ringer's lactate.
- Debride blisters and nonviable tissues.
- Use Silver sulphadiazine or Soframycin ointment to cover wounds.
- Wounds can be managed by exposure method or by closed method by covering wounds with Vaseline gauze and bandages.
- In circumferential burns in limbs and in chest **escharotomy** is performed by laterally placed incisions to prevent vascular compromise.

Nutrition (third day onwards)

- Start with oral fluids and Fortified milk
 - Calculate calories
 - Adults - $25 \text{ kcal} \times \text{Wt (kg)} + 40 \times \% \text{ burns}$
 - Children - $60 \text{ Kcal} \times \text{Wt (kg)} + 35 \times \% \text{ burns}$
- Protein requirement:
 - Adults $1 \text{ gm / kg} + 3 \text{ gm / \% burns}$
 - Children $3 \text{ gm / kg} + 1 \text{ gm / \% burns}$
- Vitamin supplements

Other supportive therapy

- Physiotherapy for the joints
- Compression dressings to prevent hypertrophic scars
- Psychotherapy
- Rehabilitation therapy

Early complications:

- Burn shock
- Renal failure

- Smoke inhalation syndrome
- Septicemia

Late complications:

- Hypertrophic scars
- Keloids
- Contractures
- Protein losing enteropathy.

DIAGNOSTIC CRITERIA, INVESTIGATIONS, TREATMENT & REFERRAL CRITERIA**LEVEL 1: AT SOLO PHYSICIAN CLINIC****Clinical Diagnosis:**

The diagnosis would be mostly clinical and history gives clue to the causative agent.

Investigations:

If facilities available, basic investigations like Hb, TLC, DLC, PCV, Urine routine examination can be carried out.

Treatment:

A solo physician can manage only minor burns at this level.

Other burns

- Assess the burn wounds
- Start IV line/ do Venesection and start fluid resuscitation
- Pain relief
- H₂-receptor antagonists
- O₂ inhalations started at 8-12 lit/min.
- Tetanus prophylaxis
- Dress the wounds
- Insert Foley's catheter, nasogastric tube if required
- Transfer patient as per guide lines / to the nearest higher centre.

LEVEL 2: AT 6-10 BEDDED PRIMARY HEALTH CENTRE**Clinical Diagnosis:**

The diagnosis would be mostly clinical and history gives clue to the causative agent.

Investigations:

- Haemogram - Hb, TLC, DLC, PCV
- Coagulation profile - BT, CT, PT
- Grouping ABO & Rh.
- Random blood sugar
- Renal function tests - urea, serum creatinine.
- Urine routine examination
- X Ray chest to detect any lung injury due to smoke inhalation.

Treatment:

For cases directly received at PHC level - they can manage only minor burns

Other burns

- Assess the burn wounds
- Start IV line/do venesections
- Start fluid resuscitation
- Pain relief
- H₂ - receptor antagonists
- O₂ inhalations started at 8-12 lit/min.
- Tetanus prophylaxis
- Dress the wounds
- Insert Foley's catheter, nasogastric tube if required
- Stabilize the patient and then transfer patient as soon as possible as per guide lines.

The cases received from solo physician-reassess the condition of patient and manage as per guide lines given above.

LEVEL 3: AT 30-100 BEDDED COMMUNITY HEALTH CENTRE

At this level moderate burns can be managed. Other cases which require transfer to a higher centre can be managed in the first two days or till they are stabilized.

Investigations: same as Level 2 **PLUS**

- Serum electrolytes
- LFT with enzymes

Treatment:

For cases received directly at this level - initial treatment is same at Level 2.

In circumferential burns in limbs and in chest, escharotomy is performed by laterally placed incisions to prevent vascular compromise and respiratory complications.

LEVEL 4: AT 100 OR MORE BEDDED DISTRICT HOSPITAL & BURN CENTRES

Can manage all types of burns including electrical and chemical burns

Investigations:

- Same as level 3
- Bronchoscopy can be carried out for smoke inhalation syndrome cases if indicated.
- Other investigations as dictated by the co-morbid condition of the patient.

Treatment:

- Treatment of these cases is as per the guide lines outlined above.
- Most of the cases received as transferred cases will be stable by the time they reach this centre, but they need reassessment and continuation of treatment.
- Specialized procedures like wound excisions and skin grafting can be carried out as per the requirement of the cases. Other modalities of treatment like physiotherapy, psychotherapy can be started in patients where indicated.

Patients with burns between 25-70% should get priority as these patients have more chance of survival compared to those with more than 70% burns.

SUGGESTED READING

1. Sabiston Text Book of Surgery, edited by CM Townsend Jr, RD Beauchamp, B Mark Evers, KL Mattox; 17th edition. 2005, published by Saunders, Elsevier, Philadelphia, USA. ISBN-13:978-81-8147-950-1.
2. Bailey& Love's Short Practices of Surgery, edited by RCG Russel, NS Williams, CJK Bulstrode; 24th edition.2004, published byArnold Publishers, London. ISBN 0 340 80820 9
3. Oxford Text book of Surgery, edited by Sir PJ Morris, WC Wood; 2nd edition.2000, published by Oxford University Press, London. ISBN 0 19262284 4.
4. The New Aird's Companion in Surgical studies, edited by KG Burnand, AE Young, J Lucas; 3rd edition. 2005, published by Elsevier, Philadelphia, USA. ISBN-0443 07 211 6.
5. ASI Text book of Surgery, edited by AA Hai, R B Shrivastava; 1st edition. 2003, published by Tata McGraw-Hill, New Delhi. ISBN 0-07-462149-1.